contradiction within the new ideas. But Bohr had to recognize that things were not as simple and as clear as he thought. Einstein did not want to relent on what for him was the key point: the notion that there is an objective reality, independent of whatever interacted with what. He refused to accept the relational aspect of the theory, the fact that things manifest themselves only through interactions. Bohr did not want to concede on the validity of the profoundly new way in which the real was conceptualized by the theory. Ultimately, Einstein accepts that the theory represents a gigantic leap forward in our understanding of the world, and that it is coherent. But he remains convinced that things could not be as strange as this theory proposed – and that, 'behind' it, there must be a further, more reasonable explanation.

A century has passed, and we are at the same point. Richard Feynman, who more than anyone has known how to juggle with the theory, has written, 'I think I can state that nobody really understands quantum mechanics.'

The equations of the theory and their consequences are used daily in a wide variety of fields: by physicists, engineers, chemists and biologists. But they remain mysterious: they do not describe physical systems but only how physical systems interact with and affect one another. What does this mean?

Physicists and philosophers continue to ask themselves what the real meaning of the theory might be and, in recent years, articles and conferences on the issue have proliferated. What is quantum theory, a century after its birth? An extraordinary dive deep into the nature of reality? A blunder that works, by chance? Part of an incomplete puzzle? Or a clue to something profound regarding the structure of the world, which we have yet to fully decipher?

The interpretation of quantum mechanics which I have presented here is the one which seems least unreasonable to me. It is called the 'relational interpretation', and it has been discussed by serious philosophers such as Bas van Fraassen, Michel Bitbol and Mauro Dorato.[5] But there is no consensus on how to think about quantum mechanics: there are other ways of thinking about it, discussed by other physicists and other philosophers. We are on the brink of that which we don't know, and opinions diverge.

Quantum mechanics is only a physics theory: perhaps tomorrow it will be corrected by an understanding of the world which is different and even more profound. Some scientists today try to iron it out a bit, to render it more in keeping with our intuition. In my opinion, its dramatic empirical success should compel us to take it seriously, and to ask ourselves not what there is to change in the theory – but rather what is limited about our intuition that makes it seem so strange to us.

I think that the obscurity of the theory is not the fault of quantum mechanics but, rather, is due to the limited capacity of our imagination. When we try to 'see' the quantum world, we are rather like moles used to living